

Robustness via Representation: Combating Distribution Shift in Imitation Learning for Robotic Manipulation

Theory & Reinforcement Learning

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1 Motivation

Imitation learning is a widely used approach for training robotic policies from expert demonstrations. However, these approaches often fail when deployed in the real world, arising from a distribution shift between the expert states the policy was trained on and the states encountered by the policy at test time, leading to compounding error. This distribution shift is particularly problematic in continuous dynamical systems like robotic manipulation. A state representation is a map of raw observation data to the policy's input space; while many exist, robotics learning benefits from mappings that prioritize structure and smoothness. This project aims to explore how different state representations can improve the robustness of a system by serving as a buffer against drift. By understanding how different latent spaces handle dynamics perturbations in addition to observational noise and perturbations to initial conditions, we can design state representations that inherently regularize the policy.

2 Methods

We focus on behavior cloning, where we use supervised learning to train a model to predict expert actions from states. We will do this using PyTorch and the Meta-World simulation environment introduced by Yu et al. (2021). The independent variable is the state representation used by the policy. While we fix the policy architecture, we vary the state encoder. As a baseline, we use the raw 39D state vector. We will compare this with other representation mappings. For example, we will use a temporal contrastive encoder to map temporally close states to nearby points in the latent space. Also, we will use a variational autoencoder (VAE) to enforce a smooth, Gaussian-distributed latent manifold through a bottleneck. The training objective will minimize the mean squared error (MSE) between the predicted and expert actions. Lastly, we will investigate a joint loss objective to create a hybrid latent space that is both discriminative (from the contrastive loss) and smooth (from the VAE loss).

2.1 Dataset

The dataset we plan to use is the Meta-World Benchmark, specifically starting with MT1. The documentation can be found [here](#).

2.2 Prior Research

In a paper by Tayal et al. (2025), researchers used VAEs to address error stemming from inaccurate measurements of environmental parameters (e.g. observational shifts). We will try using contrastive encoders, which should help address errors caused by dynamic perturbations. A paper by Nachum and Yang (2021) provides a bound between a policy learned through contrastive learning and expert policy, making temporal contrastive encoding (TCE) promising. Previous papers by Zheng et al. (2024) and Ziarko et al. (2025) have respectively implemented TCE in online/offline RL settings and temporally complex domains like Sokoban/Rubik's cube. To our knowledge, none have implemented TCE in a robotic manipulation environment. A combination of the two may also lead to better results, as each method excels in different scenarios.

3 Intended Experiments

Experiments will be conducted in Meta-World on basic tasks such as reach-v2 and push-v2. After training each model on expert data, they will be evaluated on nominal conditions and then under three forms of controlled distribution shift: observational noise, initial condition shifts in position, and

dynamics perturbations for friction or gravity constants. Performance will be measured using success rates and trajectory divergence from expert behavior. By changing perturbation magnitude, we seek to highlight differences in robustness across distinct state representations.

4 Team Contributions

- **Nancy Bao:** Lead the integration of the Meta-World simulation environment (MT1) and the collection of expert demonstration datasets; achieve baseline implementation.
- **Anastasia Lee:** Lead the development of the VAE; execute joint loss objective; develop the evaluation suite to measure success rates and trajectory divergence.
- **Jonathan Lu:** Lead the implementation of the contrastive encoder; design and execute the controlled distribution shift experiments.

References

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